Time-Domain Radiation Model of a Cap on a Rigid Sphere

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Introduction

The velocity-to-pressure exterior wave field expansion for spherically shaped radiators is well known [1–3]. It involves a modal radial filter—based on the spherical Hankel function and its derivative—which can be given as a Laplace-domain transfer function. Its frequency-dependent part can be expressed as a partial fraction expansion, which is consequently suited for the impulse invariance digital filter design. This was utilised for radial directivity control of spherical loudspeaker arrays [4], which essentially deals with the inverse filters compared to the radiation case discussed here.

Recently, a band-limited impulse invariance method (BLIIM) was introduced [5]. This improves the diffraction filters for spherical microphone array models by reduction of time-domain aliasing [6–8]. In the present paper, BLIIM is used to model radiation from a spherical loudspeaker array. For a given velocity profile of spherical cap pistons on an otherwise rigid sphere, the direction-dependent radiation and the far-field directivity of such sources can be modelled with dedicated digital filters. The proposed filter design is discussed in this paper.

Radiation Model

Conventions from [9] for the spherical harmonics $Y_{nm}(\cdot)$, the Legendre polynomial $P_n(\cdot)$ and the polar/azimuth angle pair $\boldsymbol{\theta} = [\theta, \phi]$ are used. An axial-symmetric spherical cap piston at the north-pole with unit velocity and the half-opening angle α exhibits the modal coefficients $v_{n0} = \sqrt{\frac{\pi}{2n+1}}\big(P_{n-1}(\cos\alpha) - P_{n+1}(\cos\alpha)\big)$ [1, 2, 9, 10]. Superposition of L cap pistons, facing to directions $\boldsymbol{\theta}_l = [\theta_l, \phi_l]$ and driven by frequency-independent gains g_l, yields the spectrum v_{nm} from full rotational spherical convolution [11, (82)] of $g_l\, v_{n0}$ with the band-limited modal spectrum $Y^*_{nm}(\boldsymbol{\theta}_l)$ of a Dirac impulse sifting the direction $\boldsymbol{\theta}_l$ [12, Ch. 7]. Evaluating the direction $\boldsymbol{\theta} = [\theta, \phi]$ yields the n-th order velocity's spectral coefficient, cf. [3]

$$v_n(\boldsymbol{\theta}) = \sum_{m=-n}^{+n} \underbrace{\sum_{l=1}^{L} \sqrt{\frac{4\pi}{2n+1}} g_l\, v_{n0}\, Y^*_{nm}(\boldsymbol{\theta}_l)}_{v_{nm}} Y_{nm}(\boldsymbol{\theta}). \quad (1)$$

Considering the $\mathrm{e}^{+\mathrm{j}\omega t}$ time convention[1] and the finite expansion order N, the velocity-to-pressure exterior wave field expansion from a cap-equipped, rigid sphere with radius R to radius r and direction $\boldsymbol{\theta}$ reads [1–3, 12, 13]

$$p(\omega, r, \boldsymbol{\theta}) = \rho_0 c \sum_{n=0}^{N} \underbrace{-\mathrm{j}\frac{h_n^{(2)}(\frac{\omega}{c}r)}{h_n^{'(2)}(\frac{\omega}{c}R)}}_{H_n(\omega,r,R)} \cdot v_n(\boldsymbol{\theta}), \quad r > R \quad (2)$$

with speed of sound c in m/s, density of air ρ_0 in kg/m^3, sound pressure p in Pa and modal velocity v_n in m/s. It involves the n-th order spherical Hankel function $h_n^{(2)}(\cdot)$ of second kind and its derivative $h_n^{'(2)}(\cdot)$ [14, Ch. 10.47, 10.51]. Since $H_n(\omega, r, R)$ in (2) is independent from v_{nm} and $v_n(\boldsymbol{\theta})$, this filter can be used for any given velocity profile. The frequency response $H_n(\omega, r, R)$ can be represented as Laplace-domain transfer function by $\mathrm{j}\omega \to s$ due to series expansions and recurrence relations of the involved Hankel functions [14, (10.49.7), (10.51.2)]. Therefore, (2) can be alternatively stated as

$$p(s, r, \boldsymbol{\theta}) = \rho_0 c\, R\, \frac{\mathrm{e}^{-s\cdot(\frac{r-R}{c})}}{r} \sum_{n=0}^{N} \big(1 + \hat{H}_n(s, r, R)\big) \cdot v_n(\boldsymbol{\theta}), \quad (3)$$

where the radial filter $\big(1 + \hat{H}_n(s, r, R)\big)$ features $n+1$ poles and $n+1$ zeros for the n-th order. Note the far-field amplitude decay $\propto \frac{1}{r}$, the acoustic impedance of air $Z_0 = \rho_0 c$ and the sound propagation duration $\tau = \frac{r-R}{c}$ in (3); the latter requires appropriate fractional delay filtering, which is not discussed here.

Radial Filter in Laplace Domain

The derivation for the n-th order radial filter's Laplace-domain transfer function $\big(1 + \hat{H}_n(s, r, R)\big) =$

$$\begin{cases} = \dfrac{s \cdot \sum_{k=0}^{n} \beta_n(k) \cdot (\frac{r}{c})^{k-n} \cdot s^k}{\sum_{k=0}^{n+1} \gamma_n(k) \cdot (\frac{R}{c})^{k-n-1} \cdot s^k} = 1 + \sum_{k=0}^{n} \dfrac{\mathrm{Res}_{n,k}}{s - \frac{c}{R}\mathring{s}_{\gamma_n(k)}} & (4) \\[2ex] \approx \dfrac{s \cdot s^n}{\sum_{k=0}^{n+1} \gamma_n(k) \cdot (\frac{R}{c})^{k-n-1} \cdot s^k} = 1 + \sum_{k=0}^{n} \dfrac{\mathrm{Res}^{\mathrm{Far}}_{n,k}}{s - \frac{c}{R}\mathring{s}_{\gamma_n(k)}} & (5) \end{cases}$$

is similar to the derivation in [6, 7], here differing in one additional zero in the s-plane's origin, which adds first order differentiation characteristics. This additional zero always yields a parallel, pass-through, unit-gain system (i.e. impulse response $\delta(t)$), which motivated above notation similar to [5, (1)]. The partial fraction expansion (PFE) representation of $\hat{H}_n(s, r, R)$ is exclusively used in the remainder. The coefficients $\beta_n(k)$ and $\gamma_n(k)$ follow from the mentioned series expansions of the spherical Hankel function and its derivative, respectively; cf.

[1] With this convention the standard definition of the Laplace transform is used, which asks for the kernel e^{+st} for the inverse transform. The region-of-convergence is right of the right most pole for causal signals.

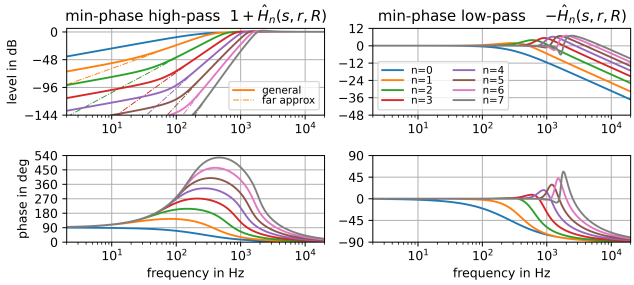


Figure 1: Radial filter design for $c = 343$ m/s, $r = 2$ m, $R = 0.2$ m, orders $n = 0 \ldots 7$. Left: the radial filter is a high-pass filter with order dependent ripples in the pass band (not visible with the chosen resolution) with a $6(n+1)$ dB/octave slope that always merges to 6 dB/octave slope and $+90°$ phase towards $\omega \to 0$ due to the single zero in the origin, cf. (4). The far-field approximation always exhibits the $6(n+1)$ dB/octave slope down to $\omega = 0$ due to $n+1$ zeros in the origin, cf. (5). Right: the polarity inverted parallel filter, i.e. $-\hat{H}_n(s,r,R)$, is a minimum-phase low-pass filter with 6 dB/octave slope and $-90°$ phase for $\omega \to \infty$, which exhibits order dependent ripples in the pass band. These determine the n-th order dependent phase excess.

[6, (2)-(6)] and [7, (3)-(6)] for explicit equations. The γ-polynomial for the n-th order exhibits $n+1$ roots (which belong to the $n+1$ poles of the radial filter), denoting the k-th root with $\mathring{s}_{\gamma_n(k)}$. For the n-th order, the k-th residual $\mathrm{Res}_{n,k}$ needs to be found such that the equality in (4) between the polynomial factorization (left) and the PFE representation (right) holds. The case in (4) considers the general filter from (2). The case in (5) is a simplification under the far-field/high-frequency assumption $h_n^{(2)}\left(\frac{\omega}{c} r \to \infty\right)$ [14, (10.52.4)] and always yields $n+1$ zeros in the s-plane's origin. Both cases represent causal, stable, minimum-phase filters.

The radiation model (3) and (4) is consistent with that of a pulsating sphere with unit-gain velocity $v_R = 1$ m/s at the sphere's radius R [15, Ch. 3.3]

$$p(\omega, r) = \frac{A}{r} \mathrm{e}^{-\mathrm{j}\frac{\omega}{c} r}, \quad A = \frac{\rho_0 c \; v_R R \; \mathrm{e}^{+\mathrm{j}\frac{\omega}{c} R}}{1 + \frac{1}{\mathrm{j}\frac{\omega}{c} R}} \tag{6}$$

(the case for the $\mathrm{e}^{-\mathrm{j}\omega t}$ time convention is discussed in [1, Ch. 6.7.8]). To validate this, $v_{nm} = 0$ for $n > 0$ is required and thus $v_{n>0}(\boldsymbol{\theta}) = 0$. Further, a north-pole cap with half-opening angle $\alpha = \pi$ spans the whole sphere. Hence, an omni-mode breathing sphere with unit-gain velocity is described with $L = 1$, $\boldsymbol{\theta}_{l=1} = [0, 0]$, $g_{l=1} = 1$ and thus $v_{n=0,m=0} = \sqrt{4\pi} \to v_{n=0}(\boldsymbol{\theta}) = \mathrm{const} = 1$. In (4) only $n = 0$ is relevant, where $\mathring{s}_{\gamma_{n=0}(k=0)} = -1$ with its residual $\mathrm{Res}_{n=0,k=0} = -\frac{c}{R}$ holds. This yields the first order high-pass (i.e. a first order low-pass is subtracted from a unit-gain pass-through system)

$$\left(1 + \hat{H}_0(s,r,R)\right) = 1 + \frac{-\frac{c}{R}}{s + \frac{c}{R}} = \frac{s}{s + \frac{c}{R}} = \frac{1}{1 + \frac{1}{\frac{s}{c} R}}, \tag{7}$$

with unit-gain pass band for angular frequencies $\omega \gg \frac{c}{R}$. This models the minimum-phase part of the frequency response in (6). Thus the model (3), (4) matches with the model (6) for the omni-mode $n = 0$.

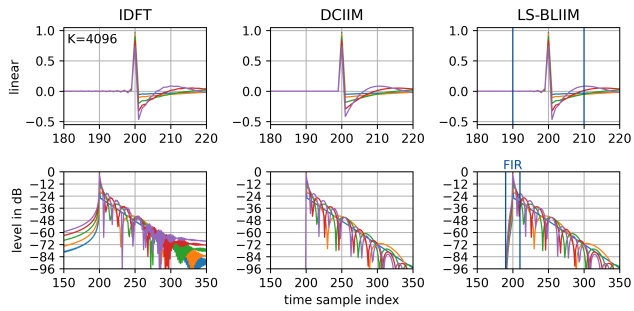
In Fig. 1, a Bode diagram exemplarily depicts the essential characteristics of the radial filter $\left(1 + \hat{H}_n(s,r,R)\right)$ and the parallel filter $\hat{H}_n(s,r,R)$, the latter being a low-pass filter with inverted polarity and the pass band's ripple complexity determined by modal order n.

Digital Filter Design

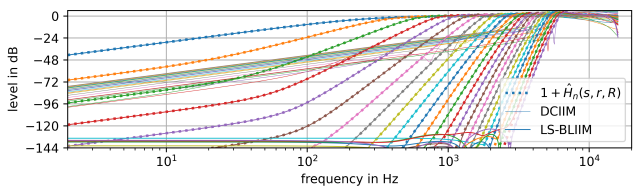
Finding the roots of the β and γ polynomials using numpy's root() and polynomial sub-package is numerically valid for about $N < 25$. The discussion here is thus restricted to these orders, although properly handling the 8 kHz octave band for a typical dodecahedron radiator with diameter 0.4 m requires about $N = 42$ following the $kR \approx N$ rule [9, p.43]. Please find numerical optimisation strategies for these higher orders in the accompanying repository[2] and upcoming paper work.

The PFE representation $\hat{H}_n(s,r,R)$ in (4), or (5), straightforwardly yields continuous-time, infinite impulse responses (IIRs) by individual inverse Laplace transform of real poles and complex-conjugate pole pairs found in the sum over k, cf. [4, 16]. Thus, the PFE is the preferred representation to use the impulse invariance method (IIM) aiming at the k-th discrete-time IIR that contributes to certain order n. Smoothing, and by that inherently band-limiting the transient of a continuous-time IIR considerably improves the standard IIM. This approach was recently introduced as band-limited impulse invariance method (BLIIM) [5]. Individually for

[2] `https://github.com/spatialaudio/daga2026_dode`



(a) Radial filters' impulse response with linear amplitude (top) and in dB (bottom) for orders $n = 0 \ldots 4$. Frequency-sampling and IDFT with block length $K = 4096$ (left), DCIIM (middle), LS-BLIIM (right).



(b) Radial filters' frequency response for orders $n = 0 \ldots 24$. DCIIM is less accurate than LS-BLIIM; both increasingly inaccurate for higher orders and closer to $f_s/2$ depending on the R/r-ratio.

Figure 2: LS-BLIIM digital-filter design for $1 + \hat{H}_n(s,r,R)$ following [7] including the sound propagation delay τ. LS-BL parameters: 21 FIR coefficients, 27 log-spaced control frequencies $[62 \ldots 16\mathrm{k}]$ Hz, $c = 343$ m/s, $f_s = 32$ kHz, $R = 0.2$ m, $r = 200 \cdot \frac{c}{f_s} + R = 2.34375$ m.

each k, the smoothing is realised with an additional, dedicated finite impulse response (FIR) filter. This FIR contains the residual between the full-band transient and the intended band-limited transient. The residual has infinite length theoretically, but can be practically windowed as it converges to zero far away from the transient. Applying the residual FIR in parallel to its corresponding IIR yields the band-limitation of the k-th full-band transient in the discrete-time domain, cf. [5, Fig. 1, (26)].

The FIR design by numerical least-squares (LS) optimisation [7] and by utilising Lagrange polynomial coefficients [8] are two comparably outperforming BLIIM approaches. For this paper, the LS approach is used due to its straightforward parametrisation and slightly more robust behaviour. The designer needs to set up an intended (comparably short) FIR length and the LS control frequencies. The source radius R (determines the location of the poles) and the sampling frequency f_s (determines the available pass band and slope of the low-pass $\hat{H}_n(s, r, R)$) are the most critical parameters. Informed heuristics leads to suitable choices, which for relaxed N typically work well at all n and k.

In Fig. 2 discrete-time IRs derived from their radial filters $\big(1 + \hat{H}_n(s, r, R)\big)$ with a) frequency domain sampling and subsequent IDFT, b) the DC matched IIM (DCIIM) and c) the LS-BLIIM are presented. Fig. 2a shows discrete-time IRs for the low orders $n = 0 \ldots 7$ and Fig. 2b frequency responses up to order $n = 24$. Note that the IRs include the sound propagation duration τ to remind that this is to be included in the complete radiation model. This inherently requires a proper pre-delay to compensate for the short group delay that is being introduced by the residual FIR filter. In the example, the IR peak exactly aligns at an integer sample index due to chosen r; in doing so no fractional delay filtering is needed here.

From Fig. 2 it can be deduced that the LS-BLIIM outperforms the IDFT approach in the time domain and the DCIIM in the frequency domain. Hence, LS-BLIIM is a reasonable trade-off strategy. The IDFT approach needs attention when aiming at correct IRs of physical models. While inherently precise in the frequency domain, it suffers from aliasing in the time domain, leading to a modified IR transient. Further, due to this time-domain aliasing, such an IR exhibits increasing energy towards the end of the DFT block, which does not match with the behaviour of physical systems. A large DFT block size and windowing after IDFT is a typical strategy to cope with the occurring pre- and post-ringing. However, larger block sizes will not undo time-domain aliasing and windowing will modify the intended frequency response. The DC matched IIM is chosen because the PFE filter $\hat{H}_n(s, r, R)$ is well defined at DC (0 dB level, ± 180 deg phase). It thus gives the best result from all potential IIM correction methods. However, the DCIIM fails to follow the target high-pass slopes at low frequencies and the pass band ripples at high frequencies (not explicitly shown here). This is due to the missing band-limitation. Hence, both, the IDFT and the DCIIM, need more care than the LS-BLIIM for the intended filter task.

Case Study: Dodecahedron Source

The regular dodecahedron with $L = 12$ cap pistons set up along its 12 faces is discussed as case study for the discrete-time radiation model of a spherical-like loudspeaker array. In the field of standardised room acoustic measurements such arrays are typically deployed with uniformly controlled loudspeaker drivers. Such a source could be utilised within an image source model [17, 18] and as theoretical time-domain model in the field of directional (room) impulse responses, cf. [3, 13, 19].

The angular positions of the loudspeaker centres are denoted by $\boldsymbol{\theta}_l$ for $l = 1 \ldots L$. The implementation in the Python package spharpy is used: polar angles are approximately at $\in \{37.38, 79.19, 100.81, 142.62\}$ deg and the azimuth is being equiangularly sampled with 60 deg, starting with $\boldsymbol{\theta}_1 = [\theta_1 = 37.38, \phi_1 = 0]$ deg. A radius $R = 0.2$ and a cap's half-opening angle $\alpha = 20$ deg reasonably models a dodecahedron source equipped with 5-inch loudspeaker drivers.

In Fig. 3 the velocity coefficients $v_n(\boldsymbol{\theta})$ for many directions $\boldsymbol{\theta}$ are evaluated by means of violin plots. The ideal regular dodecahedron with the very same gain for all cap pistons excites only certain modes n in $v_n(\boldsymbol{\theta})$ (top). Non-uniformly driven arrays, such as in the example with a dither level between ± 2 dB, will increasingly trigger all modes with increasing cap level differences (bottom).

Generally, the modal amplitude decay for increasing n is rather low. Hard truncation of the expansion is thus not recommended. Rather, a reasonable modal low-pass spectrum should be applied to $v_n(\boldsymbol{\theta})$. For that, [10, Sec. F.5] discusses a closed form solution for the spherical cap spectrum using a weighting factor α. Popular choices of directivity patterns in [10, Sec. F], [9], [20] are primarily designed for the band-limited spatial Dirac

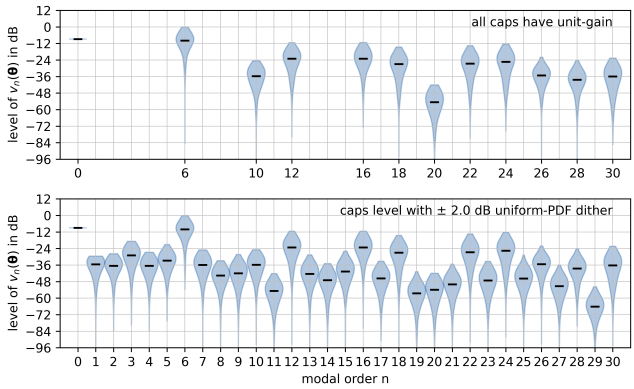
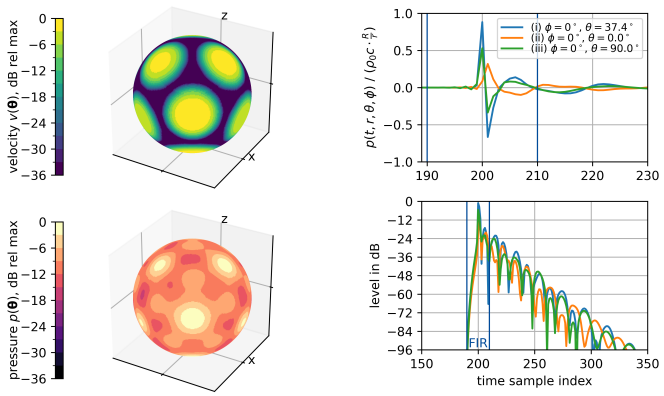


Figure 3: Dodecahedron source modelled with 12 cap pistons using $\alpha = 20$ deg. Distribution of the velocity coefficients $v_n(\boldsymbol{\theta})$ for 1059 directions $\boldsymbol{\theta}$ (spherical $t = 45$-design from Python package spharpy). Due to the symmetry of the platonic solid, the ideal regular dodecahedron excites only certain modes (0, 6, 10, 12 and all even modes from 16 up), when all caps are being uniformly driven (top). However, this is unlikely to happen with real sources due to real world variances. A simple variance model is to dither the cap levels, which triggers all other modes (bottom). The violin plots show the medians and Gaussian kernel density estimators for a chosen 0.5 dB bandwidth over the levels of the 1059 directions.



(a) Velocity profile $v(\boldsymbol{\theta})$ at radius R (top), sound pressure $p(\boldsymbol{\theta})$ in the 4 kHz third-octave band at distance r (bottom).

(b) Pressure impulse responses from (3) in the direction of a cap, the north pole and the x-axis, respectively.

Figure 4: Dodecahedron radiation model using LS-BLIIM digital filters with $N = 24$, $c = 343$ m/s, $f_\mathrm{s} = 32$ kHz, $R = 0.2$ m, $r = 200 \cdot \frac{c}{f_\mathrm{s}} + R = 2.34375$ m, 12 caps with $\alpha = 20$ deg on the faces of the regular dodecahedron, caps level with ± 2 dB uniform-PDF dither, spatially-low-pass filtered spectrum v_n using Kaiser-Bessel window with $\beta_\mathrm{KB} = 5$. FIR parameters like in Fig. 2.

impulse, not for the cap piston, but might work as well as suitable low-pass filters. In [13] a 50%-overlap Hann window was used for similar tasks. Here, the right half of the Kaiser-Bessel window is used due to its convenient parametrisation with the coefficient β_KB. For the dodecahedron under discussion, $\beta_\mathrm{KB} = 5$ seems to be a good trade-off between the aim of spatially distinct but smooth caps and the aim of properly modelling the spatial aliasing of the source for orders up to $N = 42$ to completely include the 8 kHz octave band.

In Fig. 4, results for the radiation of the modelled dodecahedron are depicted. Fig. 4a shows the velocity level at the source's radius R (top) and the resulting sound pressure level at radius r in the 4 kHz third-octave band (bottom). The applied level dither is observable, both in the velocity function as well as in the pressure pattern. The pressure pattern exhibits main lobes in the direction of the caps. For the chosen frequency band, grating lobes occur, cf. the one into direction of the north-pole and those forming the five-pointed star like shape in the front of the picture. Note that the grating lobe level is lower than the main lobe level, due to the cap directivity and the applied level dither, both acting as spatial filter for the grating lobes. Fig. 4b depicts the pressure impulse responses at radius r for three directions: (i) cap direction (here the cap at $\boldsymbol{\theta}_1$), (ii) direction of a vertex (here the north-pole), and (iii) direction into x-axis (close to the cap at $[\theta = 100.81, \phi = 0]$ deg). The IR for (i) has the largest peak as expected. As (iii) is close to a cap direction, but not exactly in the centre of the main lobe, the IR peak is lower and also the IR shape is slightly different, because different diffraction applies. The IR for (ii) exhibits the lowest peak and clearly visible a later arrival time. The three caps adjacent to the north-pole vertex mostly contribute to this interference.

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